

EE 122: Introduction to Control Systems

Problem Set #9: Advanced control design in state-space

Name: _____

Submission Date: _____

Problem 1 Solve Example 7.9 in the textbook on vectored thrust aircraft. Your work should clearly show the steps to solve the problem (the theory) and the implementation of the controller that generates the plots in Figure 7.13a and 7.13b.

Submit your code, the generated plots, and your short reflection on the results.

Grading Rubric:

- 40 points (total) for expanding out the steps from the solved example clearly and deriving the controller. Partial grading possible.
- 50 points (total) for the code implementation of the LQR controller and generating the plots. Partial grading possible.
- 10 points for the reflection on the results.